

Suraksha — Impact & Ethics

A one-page brief on real-world impact, scalability, and the honesty rules we hold ourselves to.

Real-world impact

- Targets India's dominant fraud vectors: UPI collect-request traps, digital-arrest calls, KYC-expiry phishing, advance-fee loans, and remote-access (AnyDesk) scams.
- Gives the first-time user three things they rarely have: an instant verdict in their own language, an explanation of the manipulation being used, and a concrete next step.
- Teaches the 1930 golden-hour helpline and generates a ready-to-file cybercrime.gov.in complaint - materially improving the odds of recovering money.

Scalability

- Inference runs on the user's device, so marginal compute cost is zero - it scales to millions on a free Vercel tier because the server does almost nothing.
- A portable pure-TypeScript core can be embedded as a website widget, an SMS/IVR fallback for feature phones, or a WhatsApp helper.
- Adding a language is additive (lexicon + strings + corpus), not a rewrite; the released corpus and notebook let others retrain and extend.
- The adaptive drills and shareable family-poster turn each user into a scam-aware node protecting elders - impact spreads faster than cost.

Privacy & honest engineering

- All analysis is on-device; anonymous by default; no PII collection; learning progress stays local.
- We never claim to tap a live call, and never fake bank/NPCI integration.
- We publish real metrics including the unflattering ones, and cite every helpline number and URL.
- Safety-first: when uncertain, warn; explicit OTP/PIN or APK links always floor to DANGER. A visible disclaimer says Suraksha is an assistant, not a guarantee.
- Never shame the user; accessibility (18px text, 48px targets, text-to-speech) treats low-literacy and low-vision users as first-class; Gujarati is the default, not a translation layer.

Known limitations

- Corpus is being expanded to the trilingual ~1,200-row target; current metrics are pipeline validation.
- Code-mixed / romanised text and noisy audio reduce accuracy; Web Speech is unavailable on some browsers (graceful fallback).
- A determined scammer can paraphrase around a small model; the tactic engine and a growing corpus are the mitigations, and the reproducible notebook keeps retraining cheap as scams drift.